

ASPIRE Project Description (Summer 2026)

Optics-Based Modeling and Surface Preparation of Gold Substrates: Turning UV Ozone-Induced Glass Deposition into a Quantitative Thin-Film System

Student researcher: Colter Stephens (Pre-Med)

Faculty mentors: Dr. Eric Remington (Physics) and Dr. Brian W. Gregory (Chemistry & Biochemistry)

Proposed in-residence duration: 8 weeks in residence at Samford (minimum required: 6 weeks)

Overview and Motivation

This project sits at the intersection of optics, thin-film spectroscopy, and practical surface science. Our chemistry-physics collaboration is built around reflection-based infrared spectroscopy of thin films. Dr. Eric Remington has experience with computational simulation while Dr. Brian Gregory has a thorough understanding of thin film chemistry and experimental techniques. We hope to extract quantitative information from thin films experiments and to pair that with optics-based simulation that can bolster our claims.

During UV ozone cleaning of gold surfaces (a standard procedure for removing carbonaceous contamination and stripping prior films), we have observed evidence consistent with an ultrathin glass/silica layer depositing onto gold. Rather than treating that as a nuisance, we now treat it as an opportunity to create a reproducible, nanoscale film system that we can characterize quantitatively and then leverage as a testbed for improving our modeling tools.

In parallel, we are pursuing a second, very practical goal that matters for both departments. Gold-on-glass and gold-on-mica samples that begin pristine will, over time, accumulate organics from air exposure. Many cleaning approaches roughen the gold and degrade surface quality. We therefore want to evaluate controlled furnace annealing as a route to restoring a clean, smooth gold surface. This work supports ongoing spectroscopy and directly strengthens our ability to prepare samples for the scanning tunneling microscope that Chemistry and Physics have recently obtained jointly.

This proposal has three substantial components which will be more than enough to keep a motivated student fully engaged for a full summer in residence.

Foundation from Summer 2025 (ASPIRe)

This is not a start-from-zero summer project. With ASPIRe support last summer, our team developed a working computational foundation for thin-film optics and used it alongside experimental IR data collection. The student researcher (Joe Schellenberg) built a Python-based multilayer simulation tool that can:

1. compute reflection and transmission for multilayer stacks,
2. incorporate complex refractive indices ($\tilde{n} = n + ik$),
3. include thickness-dependent phase and evanescent-field behavior,
4. generate analysis-ready plots and spectra from wavelength-dependent optical constants.

That codebase was intentionally developed to be generalizable, so that it can be applied beyond a single materials system, and it now provides a clear runway for a second ASPIRe summer. This year, Colter will not be spending weeks reinventing the wheel, but he will be extending an existing tool and applying it to an active experimental problem.

Just as importantly, the timing is ideal since Colter completed my Computational Physics course last fall and is currently in my Optics course. The research tasks are exactly the sort of work where those two courses “click” together: turning optical physics into reliable numerical predictions and then using those predictions to interpret real spectra. In fact, the simulation framework that grew out of last summer’s ASPIRe work is being folded into my optics course materials this semester, which further accelerates Colter’s ramp-up.

Project Goals and Work Plan

Aim 1: Extend the modeling pipeline to analyze glass/silica deposition on gold during UV ozone cleaning

Goal. Push the existing multilayer optics code forward so we can take experimental spectral inputs and extract physically meaningful quantities (especially film optical constants and thickness) for ultrathin films that form during UV ozone cleaning of gold.

Work plan:

- **Model development and validation.** Start with improving the pipeline with for (a) importing experimental spectral data, (b) forward simulation of candidate stacks, and (c) parameter estimation (thickness and optical-response parameters).
- **Kramers–Kronig module.** Implement a Kramers–Kronig transformation routine so we can begin with absorbance/attenuation-style spectral inputs and extract self-consistent $n(\nu)$ and $k(\nu)$ for use in the multilayer simulation. This is a natural next step in moving from “matching spectra” to “extracting material properties.”
- **Application to UV ozone sessions.** Use the model to quantify whether a silica/glass overlayer is present after cleaning, and if so, estimate thickness and optical response. The key deliverable is a workflow that makes these inferences reproducibly, not a one-off fit. This pipeline will be packaged for reuse in future student projects and in course-based demonstrations.

Aim 2: Explore whether UV ozone processing can be applied to other materials

Goal. Determine whether the same UV ozone technique produces measurable and useful thin-layer behavior on other substrates (e.g., alumina, diamond/diamond-like carbon, and other candidate materials we can access locally).

Work plan:

- Identify 2–3 candidate materials and define a small, controlled exposure matrix (time, distance, and cleaning protocol).
- Collect before/after spectral measurements.
- Use the model from Aim 1 to distinguish genuine material changes from optical artifacts due to thickness and substrate effects.

Aim 3: Regenerate gold surfaces for spectroscopy and STM sample preparation via controlled annealing

Goal. Develop and evaluate an annealing-based protocol to restore contaminated gold-on-glass and gold-on-mica samples toward a smooth, “pristine” state after air exposure, without introducing surface roughness that compromises spectroscopy or STM.

Work plan:

- Use the high-temperature furnace in my lab, with controlled-environment attachments, to anneal gold-coated substrates under a small set of conditions (temperature, time, and environment).
- Establish practical evaluation criteria (spectral indicators, surface-quality checks available to our labs, and experimental repeatability).
- Iterate toward a protocol that is reliable enough to be used routinely in support of ongoing chemistry and physics work, including STM sample preparation needs.

Student Activities and Mentoring Plan

Colter’s work will be intentionally structured so that he contributes meaningful deliverables each week while gaining research independence.

Colter will:

- Extend and test code modules (including the Kramers–Kronig workflow and analysis scripts).
- Run parameter studies and compare simulations to experimental spectra.
- Maintain a reproducible workflow (organized repository, documented notebooks/scripts, and a running lab notebook).
- Participate in the experimental context (sample preparation steps such as UV ozone cleaning, and understanding what constitutes high-quality data and controls).
- Contribute to the annealing study design and execution, and help evaluate outcomes.
- Prepare a final report and a poster-style presentation suitable for Samford Student Research Colloquium.

Mentoring structure. Colter will have regular meetings with both mentors. I will focus on computational physics, optics-based modeling, verification and debugging, and the habits of reproducible research. Dr. Gregory will provide the experimental context, materials expertise, and integration with the broader chemistry research program.

Timeline (8 weeks in residence)

1. **Week 1:** Onboarding; reproduce prior simulation results; define datasets; establish workflow and baseline assumptions.
2. **Weeks 2–3:** Implement and test the Kramers–Kronig module; validate against known/controlled inputs.

3. **Weeks 3–5:** Apply the model to UV ozone gold datasets; infer candidate film properties; iterate with experimental feedback.
4. **Weeks 5–6:** Pilot UV ozone experiments on additional materials; collect and interpret spectra with modeling support.
5. **Weeks 6–7:** Annealing study: run controlled conditions, evaluate outcomes, and refine protocol.
6. **Week 8:** Consolidate results; finalize report; prepare poster/presentation; package code/documentation for reuse.

Expected Outcomes and Dissemination

By the end of the summer we expect:

- A significantly upgraded thin-film modeling tool (including Kramers–Kronig capability integrated into the workflow).
- Quantitative analysis of the glass/silica-over-gold behavior associated with UV ozone cleaning, turning a complication into a system we can measure.
- Preliminary results on UV ozone applicability to additional materials.
- A practical annealing-based protocol that improves gold sample quality for spectroscopy and STM sample preparation.
- A presentation-ready set of results, with a clear path toward conference presentation and manuscript preparation as the work matures.